Habib University

EE-426/CE-442 Cryptography and Network Security - Spring 2026
Instructor: Farhan Khan

Time = 25 minutes **Quiz 04 SOL** **Max Points: 20**

Instructions:

i. ***<u>Smart watches, laptops, and similar electronics are strictly NOT allowed.</u>***
ii. ***<u>Answer sheets should contain all steps, working, explanations, and assumptions.</u>***
iii. Single-sided A4-sized cheat sheet is allowed.
iv. Attempt the quiz on clean papers with black/blue ink.
v. Print your name and HU ID on all sheets.
vi. Turn in your question paper along with your answer sheet(s).
vii. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	Apply the concepts from number theory and algebraic structures in understanding the design of cryptographic algorithms.	
CLO 2	Evaluate the suitability of message integrity, message authentication, and key management methods when applying in a given security scenario.	
CLO 3	Explain the setup, algorithms, and security issues in existing symmetric-key and asymmetric-key cryptosystems.	
CLO 4	Apply cryptography-based network security technologies in the design of networked information systems.	✓

Question 5a [12 points]: Choose the correct answer:

i. Which of the following process is used in IPSEC to negotiate symmetric keys securely between endpoints over an unsecured intermediate media?
   - **i. Diffie-Hellman Key Exchange**
   - ii. Advanced Encryption Standard (AES)
   - iii. Secure Hashing Algorithm (SHA)
   - iv. None of the above

ii. Hashing functions like MD5 and SHA are used in IPSEC to provide which of the following services:
   - i. Data confidentiality (privacy from eavesdropping)
   - **ii. Data Integrity (data protected from being changed during transit)**
   - iii. Securely negotiating a key over a unsecure media
   - iv. Anti-replay protection

iii. What would be a good characterization of a VPN tunnel established between a telecommuter's PC using a VPN client software and a VPN Concentrator at the HQ location?
   - **i. Remote access VPN**
   - ii. Site to site VPN
   - iii. Extranet VPN
   - iv. LAN to LAN VPN

iv. Which of the following is NOT a characteristic of a VPN?
   - i. It is a secure network
   - ii. It is deployed over a shared infrastructure
   - iii. It may use tunneling techniques
   - **iv. It does not provide any cost savings to alternate connectivity options**

v. Which of the following may be used as a terminating point for a site-to-site VPN tunnel?
   - i. Router
   - ii. VPN Firewall
   - iii. VPN Concentrator
   - **iv. All of the above**

vi. Which of the following services is not provided by AH?
   - **i. Data Confidentiality (encryption)**
   - ii. Data Origin Authentication
   - iii. Data Integrity
   - iv. Protection against Anti Replay attacks

vii. Which of the following describes the capability for a VPN terminating interface to simultaneously send IPsec protected traffic and regular unprotected traffic?
   - **i. Split tunneling**
   - ii. Load Balancing
   - iii. Firewalling
   - iv. Dual Stack tunneling

viii. Which of the following services is not provided by an IPSEC tunnel?
   - i. Data Confidentiality

ii. Origin Authentication
iii. Data Integrity
iv. Protection from Spy Ware

Question 5b [4 points]:
In Internet Key Exchange, Security Associations (SAs) are established in two phases. Why are Security Associations negotiated in two phases (Phase 1 and Phase 2) in IPsec? Explain briefly.

Answer:
In Internet Key Exchange, Security Associations (SAs) are negotiated in two phases to improve both security and efficiency.

- Phase 1: Establishes a secure and authenticated channel (ISAKMP SA) between the two peers. This phase performs mutual authentication (e.g., PSK or certificates) and sets up encryption keys to protect further communication.
- Phase 2: Uses the already secure channel from Phase 1 to negotiate IPsec SAs, which define how actual user data will be encrypted and protected.

Why two phases?

- Security: Sensitive negotiation in Phase 2 is protected because it occurs inside the encrypted tunnel created in Phase 1.
- Efficiency: Once Phase 1 is complete, multiple Phase 2 SAs can be created without repeating full authentication, reducing overhead.

Thus, in IPsec, the two-phase design ensures a secure foundation first, followed by flexible and efficient data protection setup.

Question 5c [4 points]:

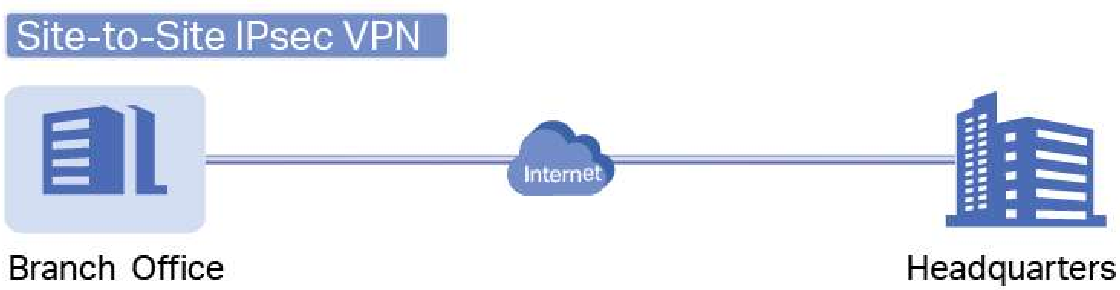

Figure 5.

Referring to the scenario as given in Figure 5 above, what typical SA Parameters will you recommend for Security Association in Phase 1 negotiation for HQ and Branch Office Routers R1 and R3?
[Hint: Remember the acronym HAGLE (Hashing, Authentication, Group, Lifetime SA, Encryption)]

Parameters	HQ R1	Branch Office R3
Key distribution method		
Encryption Algorithm		
Hash Algorithm		
Authentication Method		
Key Exchange		
IKE SA Lifetime		
ISAKMP Key		

Answer:

Parameters	HQ R1	Branch Office R3
Key distribution method	ISAKMP	ISAKMP
Encryption Algorithm	AES-256	AES-256
Hash Algorithm	SHA2-256	SHA2-256
Authentication Method	pre-share	pre-share
Key Exchange	DH Group 14	DH Group 14 (2048 bit modulus)
IKE SA Lifetime	86400	86400
ISAKMP Key	vpnpa55	vpnpa55